

Oscillations and Waves for NEET Preparation

Understand the Key Concepts of Oscillations and Waves

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Oscillation refers to the periodic back-and-forth movement between two points, such as the swinging of a pendulum or even decision-making processes. In physics, an oscillator is a device where oscillation occurs, and in the case of a pendulum, potential energy is converted into kinetic energy as it moves.

On the other hand, a wave is a physical phenomenon that transfers energy from one place to another. It is characterized by properties like amplitude, wavelength, and frequency. Waves also have a specific velocity at which they travel.

Understanding oscillations and waves is crucial for the NEET exam, and this guide will help you master these concepts with ease.

History of Inventor

Christiaan Huygens, commonly known as Christian Huyghens, was born in 1629 and passed away in 1695. He was a Dutch mathematician, astronomer, and physicist who developed the wave theory of light, revealed the exact structure of Saturn's rings, and contributed significantly to the science of dynamics. While recuperating from sickness in 1665, he became the first to record the phenomena of linked oscillation in two pendulum clocks (which he invented) in his bedroom.

Oscillatory Motion

- Oscillatory or Vibratory Motion occurs when a body moves back and forth at regular intervals, repeating its motion around a central point.
- The body moves between two extreme positions on either side of the mean position, which defines the range of its motion. Examples include the motion of a ball in a bowl, the pendulum of a clock, and a child on a swing.
- While all oscillatory motions are periodic, not all periodic motions are oscillatory.

Key Concepts of Oscillations

- **Period and Frequency**– Period refers to the smallest time interval after which oscillatory motions repeat. It can be calculated using the formula– $T =$

Frequency is the number of oscillations that occur per unit of time, represented as $V = \frac{1}{T}$, where T is the period. The unit of frequency is Hertz (Hz).

- **Amplitude**– The amplitude is the maximum displacement of the oscillating object from its mean position. It indicates the intensity of the oscillation and is represented by A .
- **Phase**– The phase helps determine the position of a particle from its mean position. It provides information on the direction and state of the motion at a given time. The initial phase refers to the state when time is zero.
- **Displacement as a Function of Time**– Displacement refers to the changes in physical quantities like position, pressure, and angle over time. It can be positive or negative and is measured as a function of time.

Linear Simple Harmonic Motion (SHM)

In linear Simple Harmonic Motion (SHM), a particle moves back and forth along a straight line. The acceleration is always proportional to the displacement from a fixed point and directed towards that point. This relationship can be expressed as $F \propto -x$ or $F = -kx$

Where–

- F is the force acting on the particle,
- x is the displacement from the equilibrium point,
- k is the force constant.

Angular SHM

In Angular Simple Harmonic Motion (SHM), the direction of angular velocity changes periodically, and the torque acting on the body is always in the opposite direction to the angular displacement. The magnitude of the torque is directly proportional to the angular displacement.

is called torsional rigidity

Or — —

Characteristics of SHM-

- Each cycle period is constant.
- Repetitious motions through a central equilibrium point.
- The motion is caused by the force directed towards the equilibrium point.
- Equipose of maximum displacement.
- is proportional to the displacement from the equilibrium.
- Acceleration =

The time reliance on a single dynamic variable that satisfies this differential equation-

. This aforementioned equation can be written in several ways-

-
-
-

Free, Damped, Forced SHM, and Resonance

- **Free Oscillation-**

This refers to motion where no external force is applied. It's also known as natural frequency. The equation for Free Simple Harmonic Motion (SHM) is-

$$F = -kx$$
, where k is a constant.

- **Damped Oscillation-** This occurs when the amplitude of the oscillation decreases over time due to resistive forces, and eventually, the system comes to rest at its equilibrium position.

Waves

When a disturbance moves from one point in a medium to another, there is no actual movement of particles. This disturbance can include electric potential, temperature, or magnetic intensity. For the disturbance to travel, the medium must possess inertia and electrical properties that allow it to propagate. These factors also influence the speed of the wave. There are two main types of waves-

Mechanical Waves- These are waves that require a medium to propagate. An example of mechanical waves is sound waves.

Non-Mechanical Waves- These include electromagnetic waves that do not require a medium for propagation. Examples include light waves, radio signals, and X-rays.

Transverse Waves

- In a transverse wave, the particles of the medium oscillate in a direction perpendicular to the direction of wave propagation.
- Transverse waves can travel through solids and along the surface of liquids.
- These waves propagate in the form of crests and troughs, with the highest point being the crest and the lowest being the trough. All electromagnetic waves are transverse.
- Examples of transverse waves include light waves, torsion waves, and water waves.

The speed of a transverse wave in solids is given by-

$$v = \sqrt{\frac{\eta}{\rho}}$$

where-

- η = modulus of rigidity of the solid
- ρ = density of the material

The speed of a transverse wave in a stretched string is given by-

$$v = \sqrt{\frac{T}{\mu}}$$

where-

- T = tension in the string
- μ = mass per unit length of the string

Longitudinal Waves-

- In longitudinal waves, the particles of the medium move in the same direction as the wave. These waves can travel through solids, liquids, and gases.
- Longitudinal waves propagate through compressions and rarefactions. When the particles are close together, it is called a compression. When the particles are spread apart, it is called a rarefaction.
- Examples of longitudinal waves include sound waves, P-wave earthquake waves, and compression waves.

The speed of longitudinal waves in solids is given by-

$$v = \sqrt{\frac{E}{\rho}}$$

The speed of longitudinal waves in liquids and gases is given by-

—

Where-

- Y = Bulk modulus of the solid
- B = Bulk modulus of the liquid or gas
- ρ = Density of the medium

Principle of Superposition of Waves

When two or more waves reach a point at the same time, the total displacement at that point is the algebraic sum of the displacements caused by each individual wave.

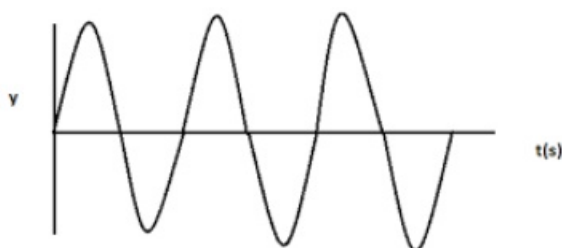
$y =$

Here, represent the displacements caused by the individual waves, and y is the overall displacement at the point.

Beats

When two waves with different frequencies travel in the same direction along the same path, the resulting amplitude alternates between maximum and minimum at different points in the medium. This alternating variation in sound intensity is known as "beats."

The frequency of beats per second is given by-



Here, the beat frequency represents the difference in frequencies of the two interfering waves.

Reflection of Waves

- When a mechanical wave reflects and refracts at a boundary, it follows the laws of reflection and refraction.
- A sound wave, when reflected from a denser medium, undergoes a phase reversal of π . However, the nature of the wave remains unchanged.
- When a sound wave is reflected from a rarer medium, the wave's nature is altered. However, there is no phase reversal.

Some Practise MCQs for NEET preparation

1. A body executes simple harmonic motion. The time period of oscillation depends on-

- (A) amplitude only
- (B) frequency only
- (C) amplitude and frequency
- (D) mass and restoring force

Answer- (D) mass and restoring force

2 The time period of a simple pendulum depends on-

2. The time period of a simple pendulum depends on

- (A) mass of the bob
- (B) length of the pendulum
- (C) amplitude of oscillation
- (D) acceleration due to gravity

Answer- (B) length of the pendulum

3. Which of the following is not a characteristic of a transverse wave?

- (A) Particles move in a direction perpendicular to the direction of wave propagation
- (B) It can travel through both solids and liquids
- (C) It is characterized by crests and troughs
- (D) All electromagnetic waves are transverse waves

Answer- (B) It can travel through both solids and liquids

4. In an oscillating system, the restoring force is directly proportional to the displacement from equilibrium. This type of motion is called-

- (A) Periodic motion
- (B) Simple harmonic motion
- (C) Uniform circular motion
- (D) Vibrational motion

Answer- (B) Simple harmonic motion

5. Which of the following waves is an example of a longitudinal wave?

- (A) Light waves
- (B) Water waves
- (C) Sound waves
- (D) Radio waves

Answer- (C) Sound waves

6. The velocity of sound in air is 343 m/s. What is the wavelength of a sound wave with a frequency of 343 Hz?

- (A) 1 m
- (B) 0.5 m
- (C) 2 m

(D) 0.2 m

Answer- (A) 1 m

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